

“DEEPSEEK IS AN EXCELLENT AI ADVANCEMENT AND A PERFECT EXAMPLE OF TEST-TIME SCALING. DEEPSEEK’S WORK ILLUSTRATES HOW NEW MODELS CAN BE CREATED USING THAT TECHNIQUE, LEVERAGING WIDELY-AVAILABLE MODELS AND COMPUTE THAT IS FULLY EXPORT CONTROL COMPLIANT. INFERENCE REQUIRES SIGNIFICANT NUMBERS OF NVIDIA GPUS AND HIGH-PERFORMANCE NETWORKING. WE NOW HAVE THREE SCALING LAWS—PRE-TRAINING AND POST-TRAINING, WHICH CONTINUE, AND NEW TEST-TIME SCALING.”



— NVIDIA STATEMENT

expert / sub-model. Of course, some jack-of-all-trades sub-models handle queries that cannot be properly classified and routed. According to a technical report published by the DeepSeek team, although the DeepSeek-V3 model has 671 billion parameters, only 37 billion are activated for any given token.

• **Inference-time compute scaling.**

DeepSeek uses a load-balancing technique that dynamically scales computing up or down to match the complexity of an assigned task.

• **Mixed-precision framework.** In AI, weights are numbers determining the strength and direction of connections between neurons in artificial neural networks. They determine the model’s ability to learn and make predictions. Instead of using 32-bit floating point numbers (FP32) for all the weights, DeepSeek has trained parts of its model with less precise 8-bit numbers (FP8), switching to 32-bit for more complex calculations where accuracy matters. This not only reduced memory usage by 75% but also expedited training.

• **Optimisation of training pipeline.** From data loading and parallelisation strategies to memory optimisation, DeepSeek has optimised every aspect of training to achieve high efficiency.

• **Multi-token prediction.** R1 predicts multiple tokens simultaneously, unlike traditional models that generate responses one token at

a time. That is, it can read whole phrases at once. This achieves the task twice as fast, with 90% accuracy.

• **Reinforcement learning (RL).**

DeepSeek-R1 uses reinforcement learning with supervised fine-tuning. Reinforcement learning is a paradigm in which a model learns by itself, without being trained with human-labelled outcomes. Like a baby learns to walk or talk, R1 also explores different ways to approach a query, picks the best one, maximises rewards for efficiency and accuracy, eliminates wrong approaches, and so on. The previous version of DeepSeek’s chain-of-thought reasoning model, DeepSeek-R1-Zero, was able to learn problem-solving strategies using large-scale RL without supervised fine-tuning. Still, it exhibited some problems like poor readability and language mixing. DeepSeek-R1 uses multi-stage training and cold-start data (a curated set of high-quality, detailed chain-of-thought examples) prior to RL to overcome that.

• **Model distillation.** People learn through experience—the

longer they work in a field, the better they can handle everyday tasks and exceptions. That is why experienced personnel are often tasked with teaching trainees the nuances of how things work in a company, right? Well, that is what model distillation is all about. Larger models learn better through RL than smaller ones. However, there are some resource-constrained environments where you cannot run a large model. So, model distillation makes a larger model teach the reasoning patterns it has learnt through RL to smaller and more efficient models, essentially transferring the knowledge and capabilities of the 671-billion-parameter model into more compact architectures. DeepSeek has distilled six models from R1, using Llama and Qwen architectures, ranging from 1.5-70 billion parameters. Although there are some capability trade-offs, distilled models have better inference speeds and lower computational costs. Experts point out that a distilled model can sometimes outperform the foundational model, like a student surpasses the teacher!

Affordable and open source

DeepSeek provides free access to its models through its web platform and mobile applications. The DeepSeek AI Assistant can be downloaded and used from the Google Play and Apple App Store for free. Within a few days of its release, the app overtook ChatGPT on Apple’s App Store.

On the other hand, DeepSeek also offers a token-based

“JEVONS PARADOX STRIKES AGAIN! AS AI GETS MORE EFFICIENT AND ACCESSIBLE, WE WILL SEE ITS USE SKYROCKET, TURNING IT INTO A COMMODITY WE JUST CAN’T GET ENOUGH OF.”

— SATYA NADELLA, CHAIRMAN AND CEO, MICROSOFT

